



國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

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COURSE: CSIC30100 BRAIN-COMPUTER INTERFACE

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Homework 2: EEG Analysis

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1 Introduction

2 Multiple Choice

This section provides the answers and concise justifications for the multiple-choice questions.

2.1 Problem 1: Signal Averaging

Answer: (C) 16

Justification: The measured signal in a single trial k is $z_k(t) = s(t) + n_k(t)$, where $s(t)$ is the deterministic signal and $n_k(t)$ are i.i.d. random noise processes.

The single-trial Signal-to-Noise Ratio (SNR) is:

$$\text{SNR}_{\text{single}} = \frac{|s(t)|}{\sigma(n(t))}$$

When averaging over N trials, the signal amplitude remains $|s(t)|$, while the standard deviation of the averaged noise is reduced by a factor of $\sqrt{N}$. Therefore, the SNR for the averaged waveform, SNR_N , is derived as follows:

$$\begin{aligned} \text{SNR}_N &= \frac{\left| \frac{1}{N} \sum_{k=1}^N s(t) \right|}{\sigma\left(\frac{1}{N} \sum_{k=1}^N n_k(t)\right)} \\ &= \frac{|s(t)|}{\frac{1}{N} \sqrt{N} \sigma(n(t))} \\ &= \frac{|s(t)|}{\frac{1}{\sqrt{N}} \sigma(n(t))} \quad (\text{since signal is constant and noise variances add}) \\ &= \sqrt{N} \cdot \frac{|s(t)|}{\sigma(n(t))} \\ &= \sqrt{N} \cdot \text{SNR}_{\text{single}} \end{aligned}$$

Plugging in the given values, where $\text{SNR}_{\text{single}} = 5/10 = 0.5$ and the target $\text{SNR}_N = 2$:

$$2 = |\sqrt{N}| \cdot 0.5 \implies |\sqrt{N}| = 4 \implies N = 16$$

Thus, 16 trials are required. So we need $N = 16$ trials to reach the target SNR.

2.2 Problem 2: Unsupervised Learning Techniques

Answer: (A) PCA, (D) ICA, (E) K-means clustering

Justification: Unsupervised techniques find patterns in data without using pre-defined labels.

- (A) PCA, (D) ICA, and (E) K-means are unsupervised because they do not require labeled data to identify patterns or structures.
- (B) LDA and (C) CSP are supervised because they require labels to learn how to distinguish between different classes of data.

2.3 Problem 3: Number of ICA Components

Answer: (C) The same as the number of EEG channels & (E) As many as the number of independent brain sources

Justification: The standard Independent Component Analysis (ICA) algorithm is designed to separate a signal into a number of independent components that is equal to the number of input sensors (channels). It operates under the assumption that the number of sources equals the number of measurements. Therefore, for an EEG dataset with M channels, ICA will produce M components.

3 Programming problems

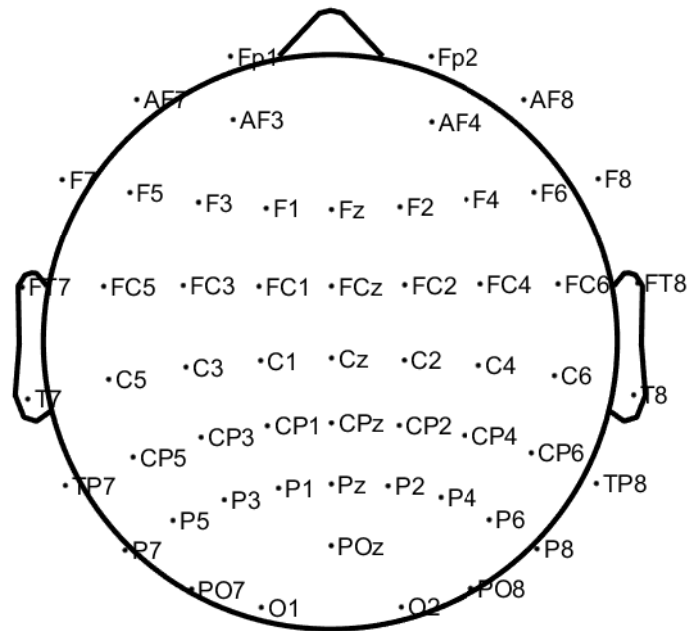
Note that for each problem, the ICA results are kept in a local file to avoid recomputation. To follow the requested format, the values are not included in the ZIP archive for this report but are available on my GitHub repository: [pb-pub/BCI-dataAnalysis](https://github.com/pb-pub/BCI-dataAnalysis).

3.1 Problem 1

3.1.1 Problem 1.1

Channel Location Map The first step of the problem is to represent the channel location map.

Problem 1-1: 2D Channel Location Map for Dataset 1



56 of 56 electrode locations shown

Click on electrodes to toggle name/number

Figure 1: Channel location map for Dataset 2

ICA computation and artifact removal Now we can perform ICA. On my personal laptop the ICA computation time is was 237.28s.

Following a visual inspection of the topographic maps for all 56 independent components, a subset was identified as artifactual and targeted for rejection. The classification of these components, based on their spatial characteristics, is detailed below:

Ocular Artifacts: Components representing electrooculographic (EOG) activity.

- **Eye Blinks (ICs 48, 56):** These components are characterized by a strong, positive, unipolar source projection focused over the pre-frontal electrode sites. This topography is a canonical representation of EOG activity generated by eye blinks.
- **Saccadic Eye Movements (IC 53):** This component projects a distinct horizontal dipole across the frontal scalp, a classic signature of lateral eye movements (saccades).

Myogenic (EMG) Artifacts: Components reflecting high-frequency muscle activity.

- **Cervical (Neck) Musculature (ICs 12, 14, 15, 18, 19, 20, 22, 23, 25, 29, 31, 37, 42, 43, 44, 45, 49):** This large class of components is defined by highly

focal, high-power activations localized to the inferior-posterior scalp periphery. These topographies are inconsistent with a cortical dipole source and are characteristic of EMG contamination from cervical muscles.

- **Temporal Muscle (IC 46):** Characterized by an isolated, focal projection over the right temporal electrode sites, this component is identified as EMG originating from temporal or mandibular muscles.

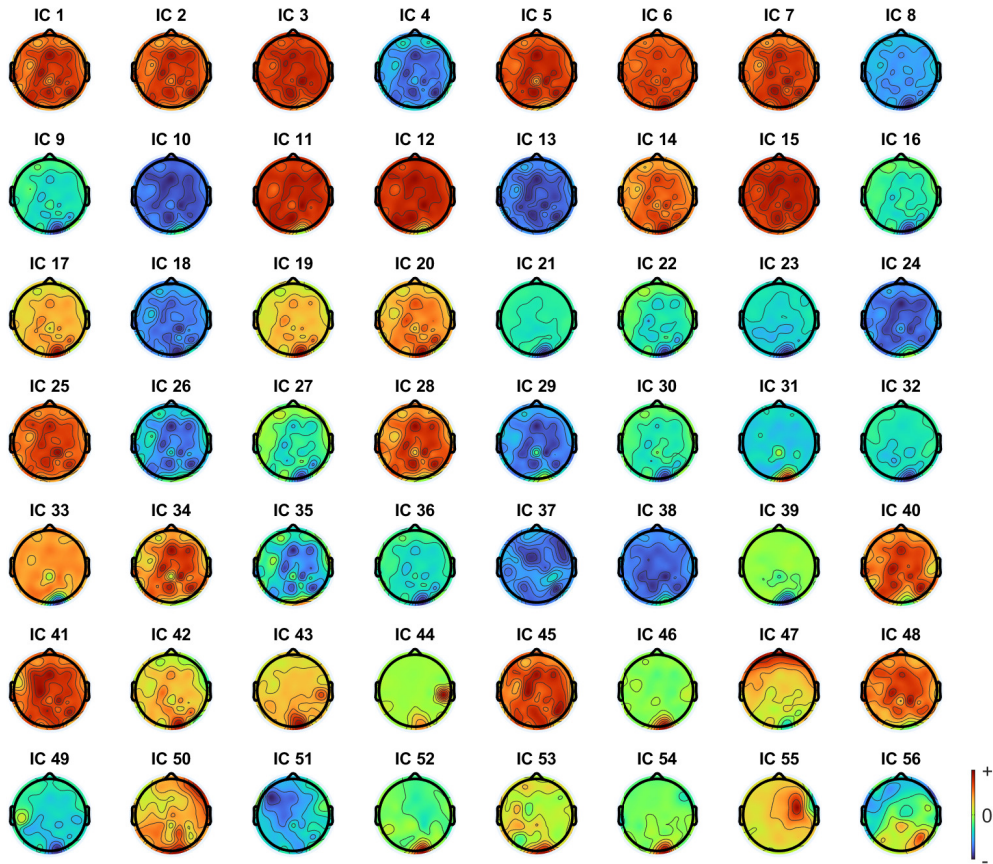
Technical and Instrumental Artifacts: • **Channel Noise (IC 50):** The topography of this component is dominated by activity in a single channel at the scalp periphery. This pattern indicates that the component has isolated noise from a high-impedance or otherwise faulty electrode.

- **Mixed-Source Artifact (IC 55):** This component lacks a physiologically plausible topography, presenting as a composite of multiple, spatially distinct artifactual sources (likely myogenic).

Residual Non-Cortical Drifts and Noise: • **ICs 2, 3, 4, 8, 11, 16, 17:** This group of components is characterized by diffuse, low-frequency, and largely monopolar scalp projections that lack any clear dipolar structure. They are interpreted as representing non-stationarities in the data, such as slow potential drifts or other residual background noise, and do not reflect organized cortical activity.

As seen above we suppress a lot of components (30/56) to clean the data. This is necessary to enhance the quality of the EEG signals. However it demonstrates the challenge of working with EEG data, where a significant portion of the recorded signals can be contaminated by various artifacts.

Figure 2: Component map with identified artifact components highlighted



Problem 1-1: ICA Component Maps

Time series analysis After removing the artifact components identified above we take a look at the time signal. We can see in Figure 3 the raw EEG time series before ICA-based artifact removal, which shows significant high-amplitude fluctuations synchronized on all channels. These can be attributed to ocular and myogenic artifacts. After artifact removal, as shown in Figure 4, the EEG time series appears cleaner. We do not see those large synchronized fluctuations anymore which suggests a successful removal of major artifacts. We can still see a high synchronization between channels, which is expected as the underlying neural signals are correlated across electrodes.

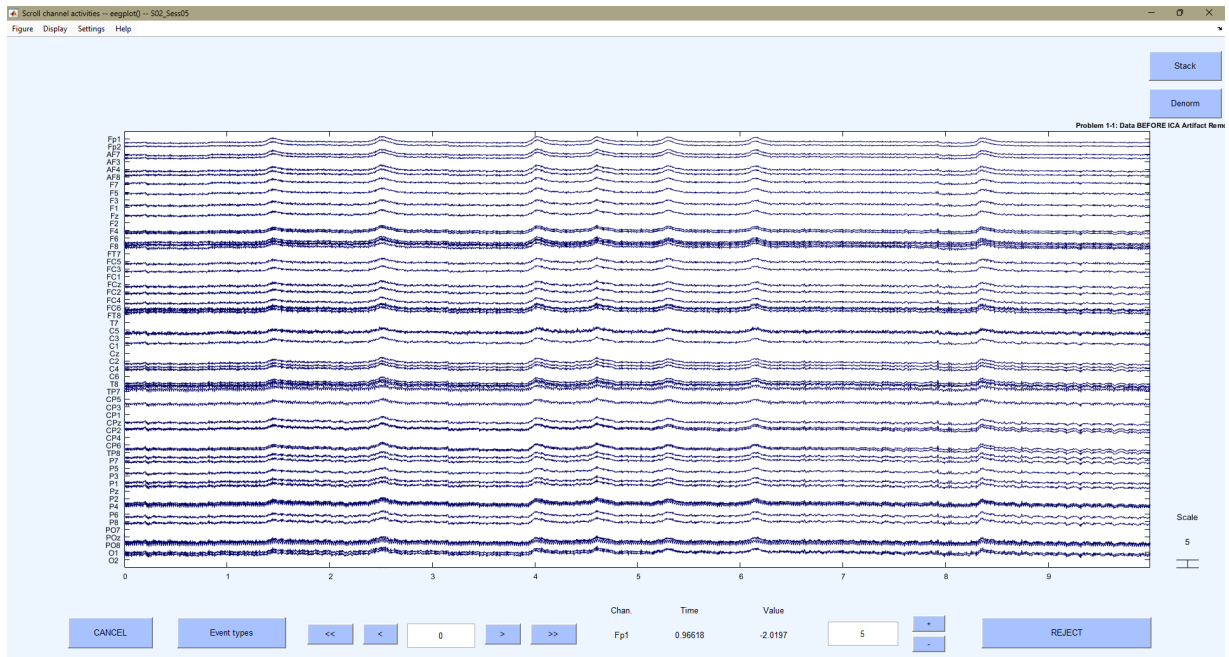


Figure 3: Raw EEG time series before ICA-based artifact removal

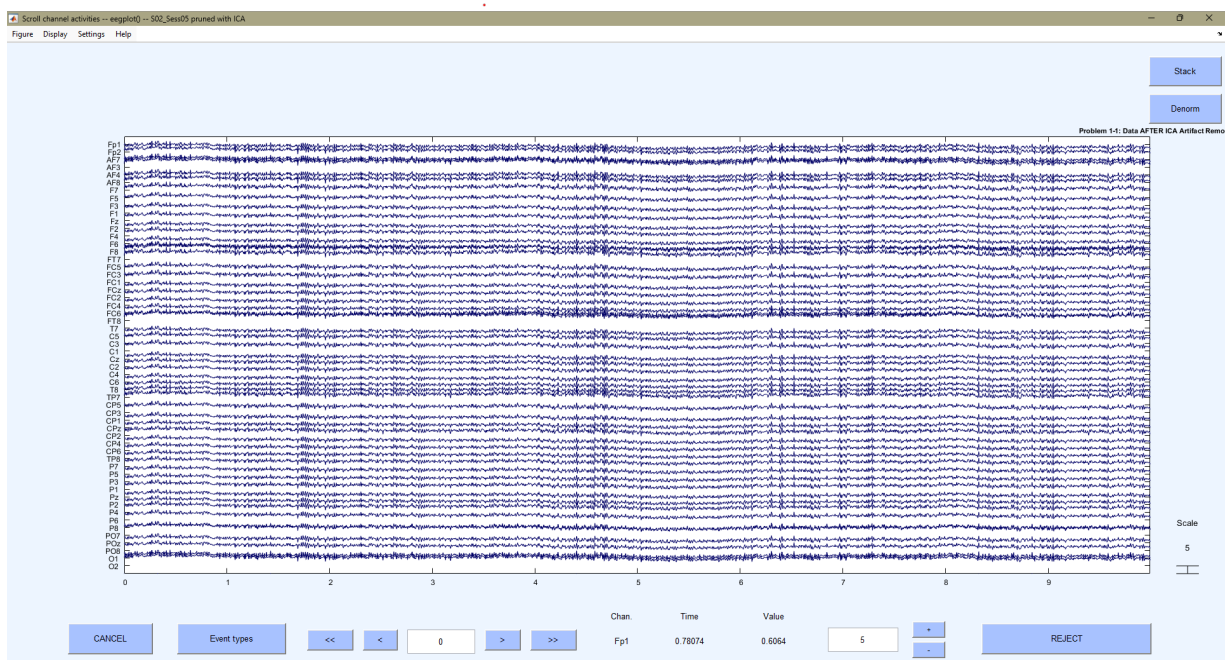
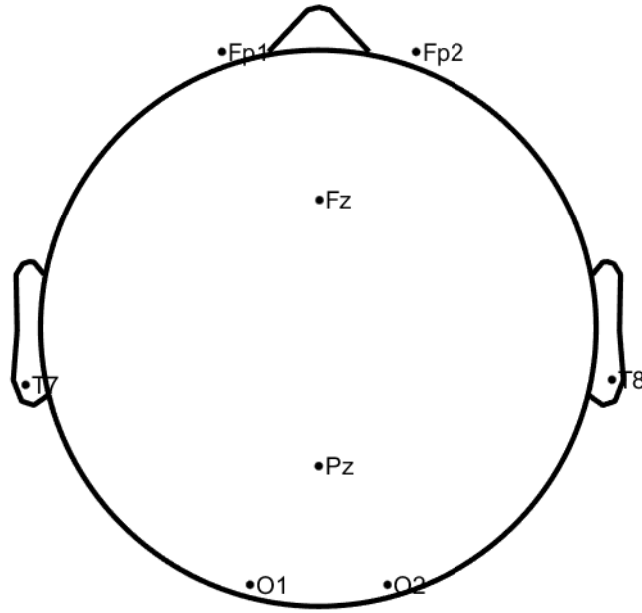


Figure 4: Cleaned EEG time series after ICA-based artifact removal

3.1.2 Problem 1.2

Location map and spectra analysis The first step of problem 1.2 is to represent the channel location map.

Problem 1-2: 2D Channel Location Map for Dataset 2



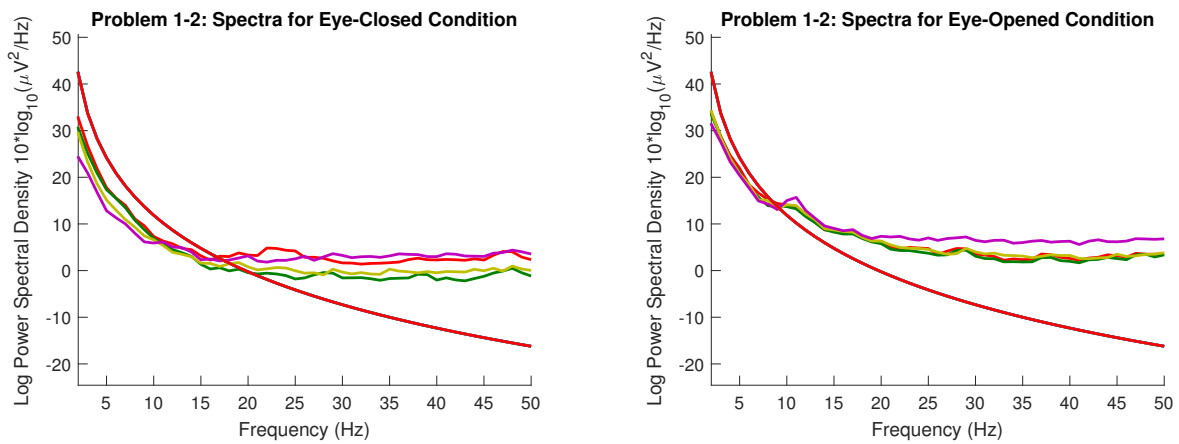
8 of 8 electrode locations shown

Click on electrodes to toggle name/number

Figure 5: Channel location map for Dataset 2

We see here 8 channels while only 4 were really used. The used ones are O1, O2, Fp1 and Fp2.

Next, we have a look at the spectra and map in 2D.



(a) Spectra and topographic maps — eye-closed

(b) Spectra and topographic maps — eye-opened

Figure 6: Spectra and topographic maps for Dataset 2: left = eye-closed, right = eye-opened.

Time series analysis Finally we can look at the time series for both eye conditions. In figure 7 we can see the EEG time series for the eye-closed condition. We can observe a higher amplitude, and in particular a high activity at the start. This one might be due to the subject adjusting

to the condition. However the rest of the signal shows more activity compared to the eye-opened condition in figure 8. Surprisingly the eye-opened condition does not show visible blinking artifacts. Overall the eye-closed condition shows more activity which is expected as closing the eyes typically increases alpha wave activity in the EEG.

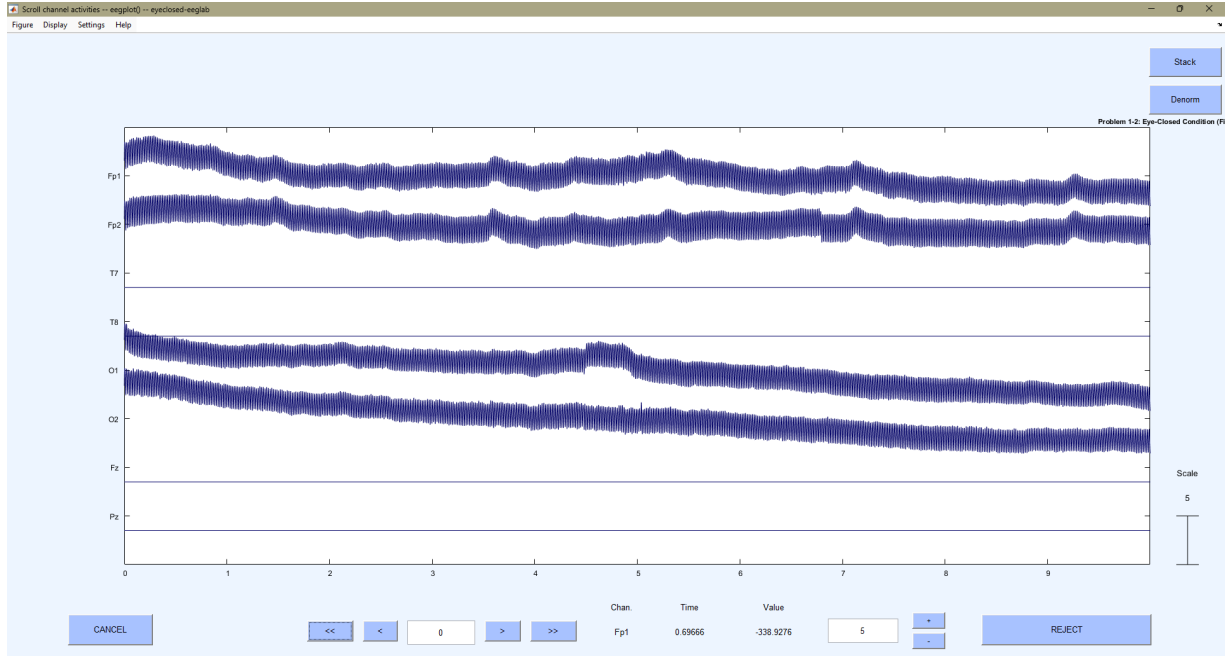


Figure 7: EEG time series for eye-closed condition

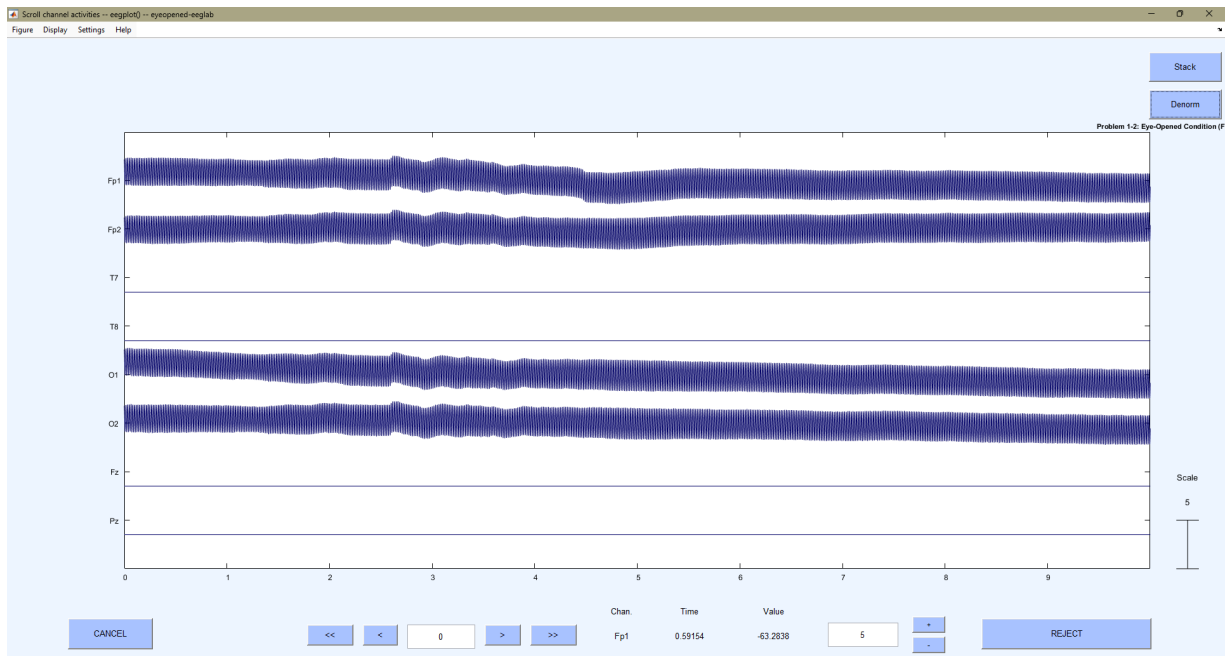
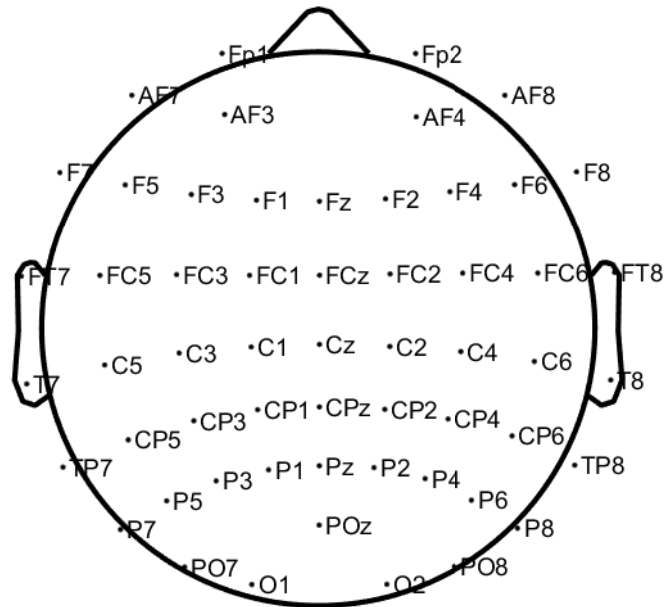


Figure 8: EEG time series for eye-opened condition

3.2 Problem 2

Channel Location Map and ICA-based Artifact Removal Same as before we start by representing the channel location map.

Problem 2: 2D Channel Location Map for Dataset 1



56 of 56 electrode locations shown

Click on electrodes to toggle name/number

Figure 9: Channel location map for Dataset 1

Next we run ICA after bandpass filtering the data between 1 and 48 Hz. The ICA computation time on my personal laptop was 402.27s which is longer than for problem 1.1. This is likely due to the increased data complexity after filtering.

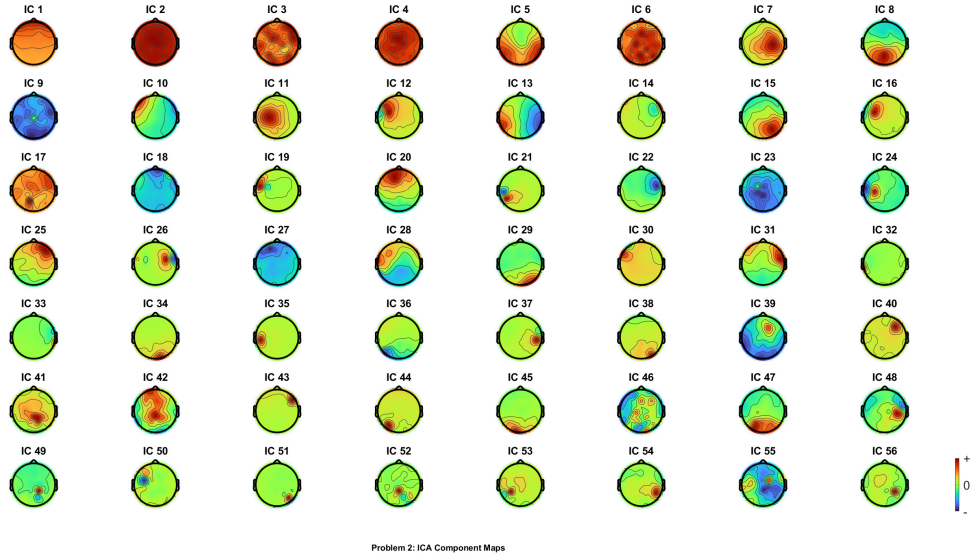


Figure 10: ICA component maps for Dataset 1

After visual inspection of the ICA component maps, a total of 13 components were identified as artifactual and marked for removal. The classification of these components is as follows:

Blinking : Component 1 exhibit strong frontal positivity, characteristic of eye blink artifacts.

Component 5 shows a vertical dipole pattern also typical of blink-related activity.

Residual Non-Cortical Drifts and Noise : Components 2, 3, 4 and 6 show diffuse, non-dipolar patterns indicative of non-physiological noise.

Myogenic artifacts : Components 17, 32, 33, 35, 43, 44, 45

s

We can note here that fewer components were removed compared to problem 1.1 (13 vs 30). This is likely due to the bandpass filtering step which helps reduce low-frequency drifts and high-frequency noise, making the ICA decomposition cleaner and reducing the number of artifactual components. The number of myogenic components removed is way lower too (7 vs 17) which suggests that the bandpass filter effectively attenuated muscle artifacts. The non-cortical drifts were also reduced (4 vs 7) likely due to the low-pass filtering removing powerline noise (50 Hz). Overall the bandpass filtering step seems to have improved the ICA decomposition and reduced the number of components needing removal.

Time series analysis After removing the identified artifact components we can look at the time series before and after artifact removal. In figure 11 we can see the raw EEG time series before

ICA-based artifact removal, which shows significant high-amplitude fluctuations synchronized on all channels. These can be attributed to ocular and myogenic artifacts. After artifact removal, as shown in figure 12, the EEG time series appears cleaner. We do not see those large synchronized fluctuations anymore which suggests a successful removal of major artifacts. We can also see a large reduction of the synchronization between channels, indicating that common noise sources have been effectively attenuated. The signals now appear more independent across electrodes, which is desirable for accurate EEG analysis.

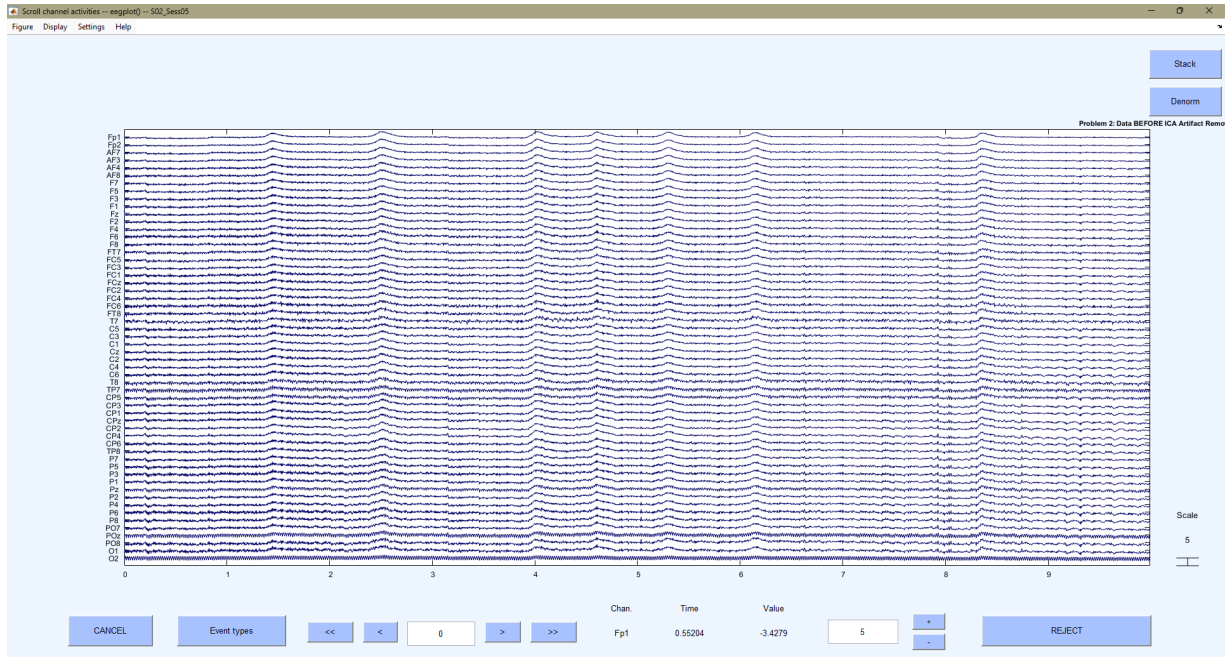


Figure 11: Raw EEG time series before ICA-based artifact removal

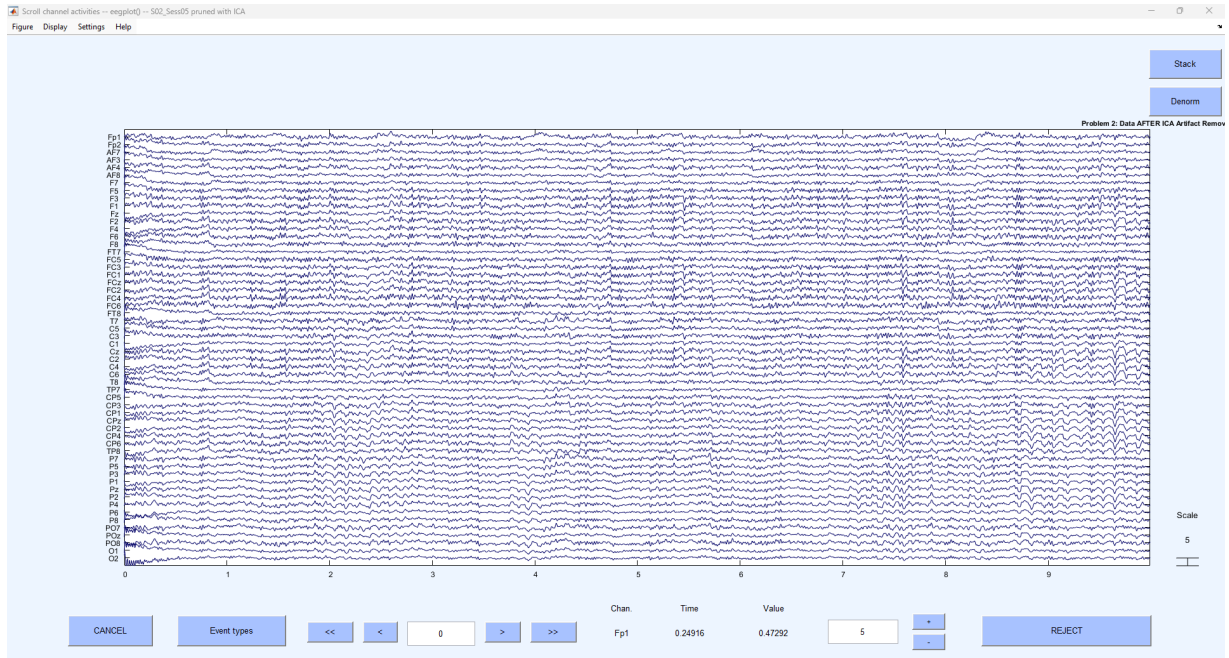


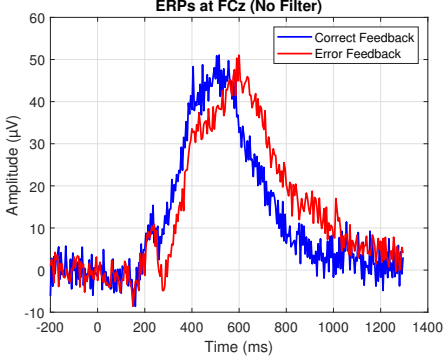
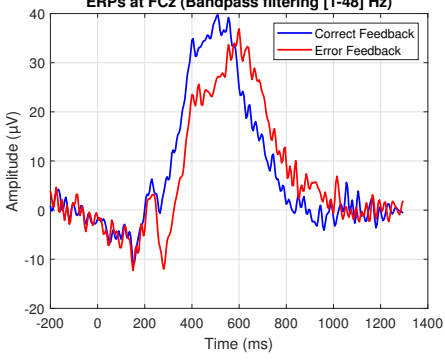
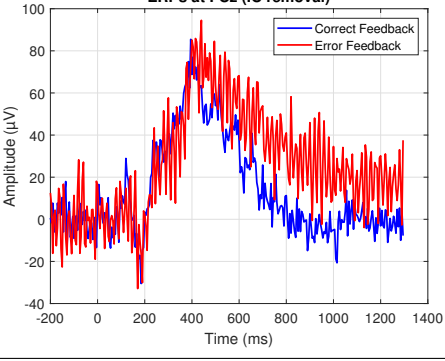
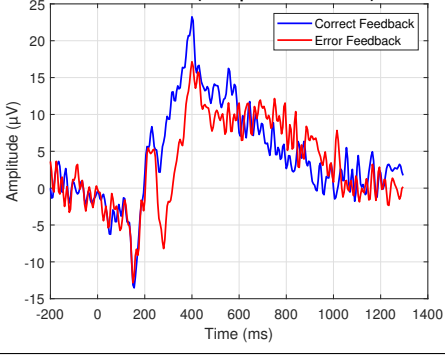
Figure 12: Cleaned EEG time series after ICA-based artifact removal

Bandpass filtering VS no filtering To compare the effects of bandpass filtering prior to ICA, we can analyze the time series plots from both Problem 1.1 (no filtering) and Problem 2 (with bandpass filtering). Our previous conclusions help us to see that bandpass filtering prior to ICA significantly improves the quality of the EEG data. The time series after artifact removal in Problem 2 shows a cleaner signal with reduced artifacts and less inter-channel correlation compared to Problem 1.1. This demonstrates the important existence of high-frequency noise and low-frequency drifts in EEG data. However the computation time for ICA is significantly increased when bandpass filtering is applied. We observed a nearly doubled computation time (402.27s vs 237.28s) on my personal laptop.

3.3 Problem 3

Methods comparison The comparison of the three methods is summarized in Table 1.

Table 1: Comparison of Preprocessing Methods for ERP Analysis

Preprocessing Method	ERP plot for 2 types of feedback	SNR (Error Feedback Only)
Without any operation		21.5184
Bandpass only		17.1671
IC removal only		7.1514
Bandpass+IC removal		9.3705

Surprisingly the best SNR is obtained without any preprocessing (21.5184). While the bandpass-

filtered data shows a lower SNR (17.1671), they still appear relatively cleaner than the raw data. We also see that ICA-based artifact removal alone significantly reduces the SNR (7.1514), while combining bandpass filtering with ICA improves it slightly (9.3705). The ICA-based artifact removal using ICLabel seems to remove a significant amount of signal along with the artifacts, leading to a lower SNR. The chosen thresholds for artifact component removal might be too aggressive, resulting in the loss of relevant neural signals. These thresholds are :

- Muscle: > 0.7
- Eye: > 0.7
- Channel Noise: > 0.7
- Line Noise: > 0.7
- Brain: < 0.01

These criteria does not seem optimal as they lead to a significant SNR reduction after ICA-based artifact removal. However they do not seem too aggressive as only 15 components were removed without bandpass filtering and 13 with bandpass filtering. It is likely that the ICA decomposition itself is not optimal, leading to components that mix neural signals with artifacts, causing loss of signal when those components are removed. A manual inspection of the components may be needed instead to optimize the artifact removal process and preserve more of the underlying neural signals.

We can also note that the computation times for the ICA without bandpass filtering was 263.32 s while with bandpass filtering it was 378.81 s on my personal laptop. It confirms again that bandpass filtering increases the ICA computation time significantly. On the ICA without bandpass filtering we removed 15 components while with bandpass filtering we removed 13 components. The bandpass filtering improvement in the ICA decomposition quality is confirmed here by the SNR increase and not anymore by the difference in number of components removed which is not significant anymore and likely due to data variability.

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